Comparison between voltage and power amplifier is given in the Table 5.1.

Table 5.1 Comparison of Voltage and Power Amplifier

SI. No.	Parameters	Voltage amplifier	Power amplifier
1.	β	High (> 100)	Low (5 to 20)
2.	R_C	High (4 – 10kΩ)	Low
3.	Coupling	Usually RC coupling	Invariably transformer coupling
4.	Input voltage	Low (a few mV)	High (2- 4V)
5.	Collector current	Low(1 mA)	High(>100mA)
6.	Power output	Low	High
7.	Output impedance	High (12kΩ)	Low(200Ω)

5.12 CLASSIFICATION OF POWER AMPLIFIERS

Transistor power amplifiers handle large signals. Many of them are driven so hard by the large input signal that the collector current is either cut-off or is in the saturation region, during the majority portion of the input cycle. Therefore, such amplifiers are generally classified according to the conduction angle produced.

1. Class A Power Amplifier
2. Class B Power Amplifier
3. *Class C Power Amplifier*

5.12.1 Class A power Amplifier

If the collector current flows at all times during the full cycle of the input signal, then the power amplifier is known as Class A power amplifier.

If the conduction angle is 360º, which means all of the input cycle is reproduced, the amplifier is called Class A power amplifier. Figure 5.26 shows the Class A power amplifier. The efficiency of Class A power amplifier is very low (less than 30%).

5.12.2. Class B Power Amplifier

If the collector current flows only during the positive half cycle of the input signal, it is called as Class B power amplifier.

If the conduction angle is 180º, which means that during the positive half cycle of the input signal the input circuit is forward biased and hence collector current flows. However, during the negative half cycle of the input signal the input circuit is reverse biased and no collector current flows. Such amplifiers are mostly used for power amplification in push-pull arrangement. In such an arrangement,

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